

Ahmed Fathy

Software Engineer

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EDUCATION

- **Cairo University, Faculty of Engineering** Giza, Egypt
 - *B.Sc. in Computer Engineering; GPA: 3.6* 2022

EXPERIENCE

- **Al Nada Scientific Office** | *Technical Support Trainee* Cairo, Egypt | June 2022 – 2024
 - **Diagnostics & Maintenance:** Troubleshoot workstations and network equipment; performed firmware updates and preventive maintenance.
 - **Documentation:** Created systematic troubleshooting documentation to streamline hardware issue resolution.

STUDENT ACTIVITIES

- **IEEE** | *UI/UX Designer - Frontend Team Leader* Cairo | Oct 2025 – Present
 - **Portfolio Platform Development:** Engineered comprehensive web platform featuring workshop and event management systems with admin dashboard for content creation, role-based access control enabling instructors to upload course materials and students to register for workshops and events.
- **Technical Center for Career Development (TCCD)** | *Frontend Developer* Cairo | June 2025 – Present
 - **Portfolio Platform Development:** Developed comprehensive platform with event pages, admin dashboard for event creation, role-based access for instructors and students, location management, event galleries, and statistical analysis pages.
 - **HR Management & Judging System:** Built HR system managing team attendance with custom form builder and reviewer functionality similar to Google Forms, alongside event judging system for evaluating and assessing teams applying to events.
- **Physics Helping District (PhD)** | *Team Leader* Cairo | June 2022 – 2024
 - **Mentorship:** Led peer tutoring initiative in electronics, achieving measurable academic improvement.
- **NASA Space Apps Challenge** | *Participant – Project: "Orrery"* Cairo | Oct 2024
 - **Orrery App:** Developed an interactive orrery app visualizing near-Earth objects using React and Three.js.

PROJECTS

- **CNN Convolution Accelerator** | *Verilog, OpenLane2, ASIC Design* Source Code | 2025
 - **Architecture:** Designed a synthesizable 2D convolution accelerator using Systolic Arrays and fully pipelined dataflow to maximize MAC throughput.
 - **Memory Hierarchy:** Implemented DMA-based loading and dual-port SRAM buffering to support up to 16x16 kernel execution.
 - **Physical Design:** Executed RTL-to-GDSII synthesis using OpenLane2, validating physical feasibility and achieving timing closure.
- **AES Encryption/Decryption Engine** | *Verilog, FPGA, Cryptography* Source Code | 2023
 - **Core Logic:** Synthesized robust AES cryptographic core in Verilog HDL supporting 128-bit, 192-bit, and 256-bit key standards.
 - **Datapath:** Engineered hardware-optimized modules for SubBytes, ShiftRows, MixColumns, and AddRoundKey transformations.
 - **Validation:** Verified cryptographic correctness against standard NIST test vectors using rigorous debug tracing.
- **Tactic (Robotics & AI)** | *Embedded Systems* Source Code | 2024
 - **System Integration:** Developed browser-controlled robotics suite integrating AI computer vision with low-latency hardware control.
 - **Firmware:** Programmed ESP32 microcontrollers to handle WebSocket communication for real-time motor control and telemetry.
 - **Backend Processing:** Deployed YOLOv5 and OpenCV on host server to process visual feedback and automate robotic decision-making.

- **Tactic (Robotics & AI)** | *Real-Time Systems, WebSockets, Embedded* Source Code | 2024
 - **Real-Time Messaging System:** Engineered a low-latency WebSocket communication layer to integrate a web dashboard with embedded hardware.
 - **Reliability:** Ensured reliable message delivery and system synchronization between the browser interface and the robotic unit.
- **Hankers (Social Platform)** | *Full-Stack Engineering* Source Code | 2025
 - **Architecture:** Built scalable social platform using Microservices principles, React.js, and TypeScript.
 - **Frontend:** Implemented responsive UI design and asynchronous API integration via AJAX.
 - **DevOps:** Established CI/CD pipelines and containerized services using Docker for consistent deployment.
- **5-Stage Pipelined Processor (RISC)** | *VHDL, QuestaSim, Computer Architecture* Source Code | 2025
 - **Core Architecture:** Engineered a 32-bit 5-stage pipelined processor in VHDL, supporting a custom ISA of over 25 instructions.
 - **Hazard Resolution:** Optimized throughput by implementing Hazard Detection and Data Forwarding units to resolve control and data dependencies.
 - **Verification:** Validated functional correctness through rigorous simulation and testbench environments in QuestaSim.
- **Hankers (Social Platform)** | *Microservices, TypeScript, Docker, CI/CD* Source Code | 2025
 - **Scalable Microservices Architecture:** Designed a distributed system using Microservices and TypeScript to decouple services and ensure scalability.
 - **DevOps Automation:** Implemented CI/CD pipelines and Docker to automate testing and reliable delivery of services.
- **NexaMart** | *E-commerce Solution, Full-Stack* Source Code | 2025
 - **Stack:** Next.js (TypeScript), Redux Toolkit, Laravel 11, and MySQL.
 - **Functionality:** Implemented secure authentication (Sanctum), shopping cart logic, real-time order processing, and administrative analytics dashboard.
- **Tayseer-Web** | *Corporate Web Solution, Next.js* Source Code | 2024
 - **Deployment:** Deployed corporate website utilizing Next.js framework and TypeScript.
 - **Performance:** Maximized performance via Server-Side Rendering (SSR) and strict static typing for reliability.
- **Company Manager App – Al Nada Scientific Office** | *ERP System, React & PostgreSQL* Source Code | 2024
 - **Tech Stack:** React.js, Redux, Express.js, Prisma, PostgreSQL, Electron for desktop deployment.
 - **Business Solution:** Developed ERP application for Al Nada Scientific Office managing customers, suppliers, inventory, and sales records.
 - **Deployment:** Packaged application with Electron for local server deployment, enabling offline usage.
- **Blog Web Application** | *Content Management System, Node.js* Source Code | 2024
 - **Core Features:** Architected content platform with CRUD capabilities, rich-text editing (Quill), and secure image handling (Multer).
- **Interactive Orrery – NASA Space Apps Hackathon** | *3D Space Simulation, React.js* Source Code | Oct 2024
 - **Hackathon Project:** Developed during NASA Space Apps Challenge 2024, simulating orbital mechanics and near-Earth object trajectories using React.js and Three.js.
- **x86 Brick Breakers Arcade** | *Assembly (x86), MASM, Low-Level Programming* Source Code | 2024
 - **Game Logic:** Developed a real-time, multiplayer arcade game using x86 Assembly and the MASM assembler.
 - **System Control:** Directly manipulated video memory for low-latency graphics rendering and managed hardware interrupts to handle real-time I/O.
- **Alien Invasion** | *Combat Simulation, C++* Source Code | 2023
 - **Simulation:** Modeled combat scenarios using custom data structures to simulate fleet mechanics and collision detection.
- **Paint For Kids** | *C++, OOP, Data Structures* Source Code | 2023

- **Application:** Built desktop drawing application with shape tools, color fills, and interactive game modes.
- **Design:** Applied OOP principles using polymorphism for extensible shape classes and Stack-based Undo/Redo system.

- **Image Restoration Pipeline** | *Deep Learning & Computer Vision, Python* Source Code | 2024
 - **Pipeline:** Constructed end-to-end restoration pipeline: Denoising, Deblurring, and Inpainting.
 - **Algorithms:** Applied Telea (FMM) inpainting and deep learning models to restore damaged historical photographs.
- **Path Planning Analysis** | *Algorithm Benchmarking, C++* Source Code | 2024
 - **Performance:** Benchmarked pathfinding algorithms (A*, Dijkstra, RRT) analyzing computational complexity and execution time.

TECHNICAL SKILLS

- **Languages:** Python (Advanced), Bash/Shell Scripting, C++, JavaScript/TypeScript, Verilog/VHDL
- **Automation & DevOps:** Linux (Fedora/RedHat), Docker, CI/CD Pipelines, Git, Make/CMake
- **Web & Frameworks:** React.js, Node.js, REST APIs
- **Domain Knowledge:** VLSI Design Flow, ASIC Verification, DRC/LVS Concepts, Static Timing Analysis